

NISHANTH SUNDARAN

Robotics Software Engineer | C++, Python, ROS2, Computer Vision

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Portfolio: nishanthsundaran.github.io GitHub: github.com/NishanthSundaran

Work permit: Germany, § 20 AufenthG (post-study residence permit); nationality: Indian



PROFILE

Robotics Software Engineer with an M.Eng. in Artificial Intelligence and a B.Eng. in Electrical and Electronics Engineering (Elektrotechnik). I develop robotics software in C++ and Python on ROS2, covering specification, implementation, integration into target systems, testing and documentation. Hands-on computer vision for robots: 3D perception with depth cameras and point clouds, obstacle detection, and segmentation (YOLOv8). Master's thesis: ROS2 and MoveIt 2 software for safe human-robot collaboration on a Universal Robots UR3e cobot, validated on real hardware. Structured and independent way of working; communicative team player.

WORK EXPERIENCE

- 09/2025 – 06/2026 Master's Thesis Researcher, Robotics Software (ROS2, C++, Python)**
Technische Hochschule Deggendorf (THD), Technologie Campus Cham, Germany
Thesis: Human-Robot Interaction with Force Feedback in Collaborative Tasks; supervisor Prof. Dr. Dmitrii Dobriborsci; defended 14.06.2026. Code:
github.com/NishanthSundaran/Human-Robot-Interaction-with-Force-Feedback-in-Collaborative-Tasks
- Developed robotics software in Python and C++ on ROS2 (Humble) and MoveIt 2 for safe human-robot collaboration on a UR3e cobot with Robotiq 2F-140 gripper.
 - Implemented a real-time hybrid position-admittance controller at 500 Hz using the UR3e force/torque sensor, plus a C++ ros2_control Cartesian admittance controller plugin.
 - Built 3D perception with an Intel RealSense D435i: YOLOv8 instance segmentation removes the human operator from the depth point cloud, which feeds an OctoMap for obstacle detection and collision-free motion planning in MoveIt 2.
 - Integrated the software into the target system (UR3e via ros2_control and the UR ROS2 driver, Robotiq gripper via URCap socket) with matching Gazebo simulation and real-hardware launch paths.
 - Specified, implemented and executed tests: 120+ logged test runs in simulation and on the real UR3e, ROS2 linter tests (flake8, pep257), and validation of ISO/TS 15066 safety functions (speed and separation monitoring, power and force limiting, hand guiding).
 - 13-state finite state machine (FSM) for assistive lift-and-place; hand-eye calibration; HSV color segmentation with depth back-projection. Documented architecture, parameters and test results; aligned requirements regularly with the supervisor.
- 12/2023 – 11/2024 Machine Learning Intern**
Energy Future World, London, United Kingdom (remote)
- Developed Python computer vision software for aerial imagery to detect and track renewable-energy infrastructure; 85% detection accuracy with OpenCV, Scikit-learn, NumPy and Pandas.
 - Built an image-processing pipeline (edge detection, contrast enhancement) and benchmarked ML models on real-world datasets; reported results to a distributed remote team.
- 03/2022 – 12/2022 Process Associate**
Amazon, Bangalore, India
- Data analysis and auditing on operational trends; contributed to automation and forecasting workflows.
- 05/2019 – 11/2021 Project Engineer**
Power And Instrumentations Ltd, Gujarat, India
- Analysed high-voltage (HV) power-grid performance for reliability; managed cost, quality and resources across project sites in coordination with site teams.

EDUCATION

- 10/2022 – 06/2026 M.Eng. Artificial Intelligence for Smart Sensors and Actuators**, Technische Hochschule Deggendorf, Germany. Focus: Robotics, Computer Vision, Machine Learning, Deep Learning, Autonomous Systems, IoT. Thesis: Human-Robot Interaction with Force Feedback (defended 14.06.2026).
- B.Eng. Electrical and Electronics Engineering** (Elektrotechnik), University of Calicut, Kerala, India. Focus: Digital Signal Processing, Digital System Design, Power Electronics, IoT, Embedded C.

TECHNICAL SKILLS

Programming	C++, Python, CMake, Bash, Git, Linux (Ubuntu), VS Code
Robotics	ROS2 (ROS 2 Humble), ros2_control, MoveIt 2, Nav2, SLAM (slam_toolbox), Gazebo, URDF/Xacro, tf2, Motion Planning (OMPL, Pilz), Control (PID, Admittance, Impedance), Universal Robots UR3e, Robotiq grippers
Computer Vision	3D Perception, Obstacle Detection, Segmentation (YOLOv8 instance segmentation, HSV color segmentation), Object Detection, Point Clouds (PointCloud2, OctoMap), Depth Cameras (Intel RealSense D435i), Hand-Eye Calibration, OpenCV
Testing	Test specification, test implementation and execution, simulation-based testing (Gazebo), real-hardware testing, linter and unit tests (ament, flake8), system integration, technical documentation
AI / ML	PyTorch, JAX / MJX, Reinforcement Learning (PPO), Imitation Learning, Vision-Language Models (VLM), Vision-Language-Action (VLA), Scikit-learn, NumPy, Pandas
Safety	ISO/TS 15066 (speed and separation monitoring, power and force limiting, hand guiding), risk assessment
Working Style	Structured and independent working, communication skills, teamwork, coordination with other departments, Agile / Scrum

LANGUAGES

English	Professional fluency, written and spoken (C1)
German	A2 (elementary), actively learning toward B1

PROJECTS

C++ Cartesian Admittance Controller for ros2_control

Open source: github.com/NishanthSundaran/nishanth_cartesian_admittance_controller

- C++ ros2_control controller plugin: Cartesian admittance law with force-adaptive variable damping, damped least-squares Jacobian pseudo-inverse and low-pass filtered force/torque input; runs at 500 Hz in controller_manager with no memory allocation in the update loop.
- Runtime-tunable parameters via generate_parameter_library, realtime-safe diagnostics via realtime_tools; CMake / ament build, launch files and README with architecture diagram and parameter documentation.

Autonomous Mobile Robot with SLAM Navigation (ROS2, Nav2)

Case study: github.com/NishanthSundaran/SLAM_nav_bot

- Four-wheeled LiDAR robot on ROS2 Humble: slam_toolbox online SLAM and a full Nav2 stack (AMCL, NavFn, DWB) with costmap-based obstacle detection; 3/3 goals reached with 0 collisions in a cluttered warehouse simulation.

Embodied Agent: VLM-driven Pick-and-Place (ROS2, MoveIt 2)

Multimodal AI: github.com/NishanthSundaran/embodied_agent

- Vision-language model grounds objects in live RGB frames; 3D localization via depth back-projection; an 8-tool agentic loop turns natural-language commands into MoveIt 2 manipulation on a UR3e with Robotiq gripper.

One Brain, Many Bodies: Humanoid Autonomy Stack in MuJoCo

Portfolio project: github.com/NishanthSundaran/one-brain-many-bodies

- Local open-source VLM (Qwen2.5-VL-3B) plans at approx. 1 Hz via tool calling with visual grounding; compliant whole-body controller executes at approx. 500 Hz; the same software drives Unitree G1 and H1 humanoids with only per-robot configuration changes; PPO balance policy holds the pelvis to 1.4 cm while the arms reach.

G1 Motion Imitation: Learning Human Gait from Motion Capture

Portfolio project: github.com/NishanthSundaran/g1-motion-imitation

- Imitation learning (PPO) on a Unitree G1 tracking LAFAN1 motion capture in JAX / MJX at approx. 30,000 steps/s on a 6 GB GPU (9x CPU baseline); gait within 2 to 7% of the human reference on speed, cadence and stride; contributed four framework fixes upstream.

Additional Projects

Self study: github.com/NishanthSundaran

- YOLOv8 object detection with audio alerts and live GUI; histogram equalization for image enhancement; ARIMA / SARIMA air-quality forecasting (over 93% test accuracy); IoT speed detection with accelerometer and GPS.

CERTIFICATIONS

09/2026	Machine Learning with Python , IBM (Coursera). Supervised and unsupervised learning, classification, regression, clustering, model evaluation, Scikit-learn. Credential: coursera.org/verify/RQQD0CVOYW12
08/2026	AI Agents and Agentic AI with Python & Generative AI , Vanderbilt University (Coursera). Tool use, function calling, LLM application development, RAG, Python. Credential: coursera.org/verify/M5M2XTC1FN50